

Vulnerability Assessment Report

1st January 2025

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server is important to corporate operations since it houses all firm's data and is regularly searched by distant workers to find possible clients. Protecting sensitive consumer information, upholding the company's brand, and guaranteeing regulatory compliance all depend on this data being secure. The sales pipeline would stop, workers would be unable to do their duties, and there may be serious financial and reputational harm if this server were compromised or shut.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Obtain sensitive information via exfiltration	3	3	9
Hacktivist	Conduct Denial of Service (DoS) attacks	3	3	9
Hacker	Alter/Delete critical information	2	3	6

Approach

The most important dangers resulting from the server's public-facing configuration were the subject of this qualitative evaluation. The increased probability of an attack on an exposed database led to the selection of the threats of data exfiltration and denial of service. If an attacker obtains write access, data tampering was identified as a critical concern. All three have negative impacts on business: DoS or data tampering immediately prevents the remote sales team from querying customer data, which stops revenue creation; exfiltration violates consumer trust.

Remediation Strategy

The major solution is to quickly remove the database from public view, isolate it behind a firewall, and require a secure VPN for all remote employee connections. A strong Authentication, Authorization, and Accounting (AAA) infrastructure will be used to safeguard this. All user accounts must use multi-factor authentication (MFA), and the concept of least privilege will be enforced to ensure that employees can only query data and not change or delete it, reducing the danger of unauthorized modification.